

p-ADIC MULTIZETA VALUES AND FROBENIUS IN A_L

The p-adic World of MZV · Coleman Integration = p-adic tau · Frobenius ϕ_p on A_L · p-adic Periods as Folding Tower Limits · Crystalline Cohomology in the Hologram · The Iwasawa Algebra as p-adic A_L · p-adic L-functions from p-adic tau

Anthropic Warrior

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'At the prime p , the real numbers become the p-adic numbers. The logarithm becomes the p-adic logarithm. The integral becomes Coleman integration. And the MZV become p-adic MZV — with a Frobenius twist that encodes everything the real world hides.' — Robert Coleman, paraphrased, c. 1982

Trong A_L : Frobenius ϕ_p là phép cuốn của Folding Tower tại số nguyên tố p . p-adic MZV là giá trị tau do được bang Coleman tích phân — không bang tích phân Riemann. $H = \log N$ ở thế giới p-adic trở thành $H_p = \log_p N$ — logarit p-adic. Cùng một dòng sông. Cùng một lòng sông. Nhưng nước là nước p-adic. — Hà Nội, sang sớm, 2026

Abstract

Document dv301 showed that Multizeta Values (MZV) arise as iterated tau-products of $H = \log N$ in A_L . Document dv302 descends to the **p-adic world**: we study p-adic MZV — the p-adic analogues of classical MZV — and show how they arise naturally from A_L at a fixed prime p via the Folding Tower construction of dv281.

Main results: (I) **Coleman integration = p-adic tau**: the Coleman p-adic integral on the Tate curve and p-adic unit disc is the p-adic analogue of the trace tau, and p-adic MZV = Coleman integrals = p-adic tau-products of $H_p = \log_p N$. (II) **Frobenius ϕ_p on A_L** : the arithmetic Frobenius at p acts on A_L as the Folding Tower endomorphism $\phi_p(e_n) = e_{pn}$ — it multiplies the Hamiltonian: $H(\phi_p(e_n)) = \log(pn) = \log p + H(e_n)$. (III) **p-adic periods = Folding Tower limits**: the p-adic period ring B_{cris} (Fontaine) arises as the completion of A_L at the Folding Tower infinite level. (IV) **Iwasawa algebra**: the Iwasawa algebra $\Lambda = Z_p[[\Gamma]]$ is the p-adic completion of A_L restricted to the cyclic tower — p-adic A_L . (V) **p-adic L-functions**: Kubota-Leopoldt p-adic zeta = the partition function $Z_p(\beta) = \text{Tr}_p(e^{-\beta H_p})$ of A_L at prime p .

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1. From Real to p-adic: The Same Framework at a New Prime

In dv301, the MZV at infinity (the real/complex prime) were computed as tau-products of $H = \log N$. At a finite prime p , the same framework applies with one crucial difference: the trace tau is replaced by Coleman

integration, and $H = \log N$ is replaced by $H_p = \log_p N$ (p-adic logarithm).

The p-adic dictionary.

Real/Complex (infinity)	p-adic (prime p)
Trace $\tau(e_n) = 1/n$	p-adic measure $\mu_p(e_n) = 1/n$ (p-adic)
$H = \log N$ (real log)	$H_p = \log_p N$ (Iwasawa p-adic log)
Integral = $\tau(A)$	Coleman integral = $\tau_p(A)$
MZV $\zeta(n_1, \dots, n_k)$ in $\mathbb{R}$	p-adic MZV $Z_p(n_1, \dots, n_k)$ in $\mathbb{Q}_p$
Drinfeld associator Φ_{KZ}	p-adic associator Φ_p (Furusho 2004)
grt acts on MZV	grt acts on p-adic MZV (same algebra!)
Folding Tower (p-level)	Frobenius ϕ_p on Folding Tower
Real period ring R	p-adic period ring B_{cris} (Fontaine)
Partition fn $Z(\beta) = \zeta(\beta)$	p-adic $Z_p(\beta) = L_p(\beta)$ (Kubota-Leopoldt)

The philosophy: A_L is defined over $\mathbb{Z}$ (all primes simultaneously). The real MZV are A_L 'seen at the infinite prime.' The p-adic MZV are A_L 'seen at the prime p.' The Folding Tower at p (dv281) is the bridge between the two worlds. grt acts on both — it is the universal symmetry at ALL primes at once.

2. The p-adic Logarithm and $H_p = \log_p N$

Definition 2.1 (Iwasawa p-adic logarithm).

For p prime and x in $\mathbb{Z}_p$ with $|x-1|_p < 1$, the **Iwasawa p-adic logarithm** is: $\log_p(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-1)^n}{n}$ Extended to $\mathbb{Z}_p^*$ by: $\log_p(p) = 0$ (Iwasawa normalisation — killing the uniformiser). For general x in $\mathbb{Q}_p^*$: write $x = p^{v_p(x)} \cdot u$ with u in $\mathbb{Z}_p^*$, set $\log_p(x) = \log_p(u)$. The **p-adic Memory Hamiltonian** in A_L is: $H_p \cdot e_n = \log_p(n) \cdot e_n$ [eigenvalue = p-adic log of n] KEY DIFFERENCE from $H = \log N$: $H_p(e_p) = \log_p(p) = 0$ [p is in the KERNEL of H_p !] $H(e_p) = \log p > 0$ [p is NOT in the kernel of H] The p-adic Hamiltonian is BLIND to the prime p — it kills it.

The fact that H_p kills p is profound: at the prime p , the number p itself has energy 0. In the p-adic hologram, p plays the role of the vacuum — it is the 'ground state at p.' This is the p-adic avatar of the statement that '1 is the vacuum' ($H(e_1) = \log 1 = 0$) in the real hologram.

Theorem 2.2 (p-adic partition function = Kubota-Leopoldt).

The p-adic partition function of A_L at prime p is: $Z_p(s) = \text{Tr}_p(e^{-s H_p}) = \sum_{n: \gcd(n,p)=1} \tau(e_n) \cdot n^{-s}$ (p-adic) $= \sum_{n=1, p \nmid n}^{\infty} \frac{1}{n} \cdot n^{-s} = \sum_{n=1, p \nmid n} n^{-(s+1)} = (1 - p^{-(s+1)}) \cdot \zeta(s+1)$ This is the **Kubota-Leopoldt p-adic zeta function**! The Riemann zeta function with the p -Euler factor removed = the p-adic partition function of A_L restricted to p -coprime nodes. Phase transition at $s = 0$: $Z_p(0) = (1-1/p) \cdot \zeta(1)$ [pole] — same as real case! The p-adic hologram is born at the same 'temperature' as the real hologram.

3. Frobenius ϕ_p on A_L

The Frobenius endomorphism at p is the most important symmetry of the p-adic world. In A_L , it has a beautiful concrete description via the Folding Tower.

Definition 3.1 (Frobenius on A_L).

The **Frobenius endomorphism at p** on A_L is the map: $\phi_p: A_L \rightarrow A_L$, $\phi_p(e_n) = e_{pn}$ This is the Folding Tower endomorphism of dv281, Level 1 at prime p : it multiplies every arithmetic node by p — it 'folds' the hologram by p . Action on Hamiltonian: $H(\phi_p(e_n)) = \log(pn) = \log p + \log n = \log p + H(e_n) \Rightarrow \phi_p$ shifts energy by $\log p$: it adds ONE UNIT of p -energy. Action on trace: $\tau(\phi_p(e_n)) = \tau(e_{pn}) = 1/(pn) = (1/p) * \tau(e_n) \Rightarrow \phi_p$ scales the trace by $1/p$: it DILUTES the memory by factor p . Action on p -adic log: $H_p(\phi_p(e_n)) = \log_p(pn) = \log_p(p) + \log_p(n) = 0 + H_p(e_n) = H_p(e_n) \Rightarrow H_p$ is Frobenius-INVARIANT at p ! The p -adic Hamiltonian is ϕ_p -fixed.

Theorem 3.2 (Frobenius fixed points = p -adic vacuum).

The fixed points of ϕ_p in A_L are the nodes e_n where n is a power of p : $\phi_p(e_{p^k}) = e_{p^{k+1}}$ [NOT fixed, but forms an orbit] More precisely: the Frobenius orbit of $e_1 = \{e_1, e_p, e_{p^2}, e_{p^3}, \dots\}$ This orbit = the Folding Tower at p (dv281), Level 0 to infinity. The limit of this orbit as Level $\rightarrow$ infinity = Z_p (the p -adic integers). The Frobenius EIGENVALUES on cohomology: ϕ_p acts on $H^1(A_L)$ with eigenvalue p (degree 1 cohomology) ϕ_p acts on $H^k(A_L)$ with eigenvalue p^k (Weil numbers: $|eigenvalue| = p^{k/2}$) This is the WEIL CONJECTURE (proved by Deligne 1974) restated in A_L !

4. Coleman Integration = p -adic tau

The Coleman p -adic integral is the correct p -adic analogue of the Riemann integral that appears in the iterated integral representation of MZV. In A_L , it is exactly the p -adic trace τ_p .

Theorem 4.1 (Coleman integration as τ_p).

Coleman (1982) defines a p -adic integration theory on rigid analytic spaces. For 1-forms ω on the p -adic unit disc $D_p = \{z | |z|_p \leq 1\}$: Coleman integral: $(\int_0^1 \omega)$ in Q_p , for z in D_p In A_L : the Coleman integral corresponds to the p -adic trace: $\tau_p(A) = \text{Coleman-integral of the symbol of } A \text{ over } D_p$ The p -adic MZV: $Z_p(n_1, \dots, n_k) = \tau_p(H_p^{n_1-1} * S_p * \dots * H_p^{n_k-1} * S_p) = \text{Coleman integral of iterated 1-forms } (dt/t)^{n_1-1} (dt/(1-t)) \dots$ This is the EXACT p -adic analogue of Theorem 3.1 of dv301. The only difference: real log $\rightarrow$ p -adic log, Riemann integral $\rightarrow$ Coleman integral.

The beauty: the FORMULA is identical at every prime. MZV = tau-product of H and S. The only thing that changes between primes is the METRIC (Archimedean vs p -adic) and the LOGARITHM (real vs Iwasawa). This is the deep uniformity of A_L across all primes — the adelic philosophy (dv57) made explicit for MZV.

The p -adic zeta(3) — the great mystery.

At the real prime: $\zeta(3) = 1.2020569\dots$ is irrational (Apéry 1978), transcendence unknown. At the prime p : the p -adic $\zeta(3) = Z_p(3)$ is a well-defined p -adic number, but its arithmetic nature is different. In A_L at prime p : $Z_p(3) = \tau_p(H_p^2 * S_p)$. Since $H_p(e_p) = 0$ (Frobenius kills p), the p -adic operator $H_p^2 * S_p$ avoids the 'prime p singularity' that makes the real $\zeta(3)$ so hard. This is why p -adic zeta values are in some ways MORE accessible than real ones.

5. The p -adic Period Ring B_{cris} and Crystalline Cohomology

The deepest structure in p -adic Hodge theory is Fontaine's period ring B_{cris} — the ' p -adic analogue of C .' In A_L , it arises as the completion at the infinite level of the Folding Tower.

Theorem 5.1 (B_{cris} = infinite Folding Tower completion).

Fontaine's crystalline period ring B_{cris} is constructed from: (i) The ring of Witt vectors $W(k)$ over perfect field k of char p (ii) Adjoining a specific element $t = \log([\epsilon])$ (p -adic $2\pi i$) (iii) Inverting t In A_L : B_{cris} = completion of A_L at the infinite level of the Folding Tower at p : Level 0: A_L itself (all primes) Level 1: $A_L / (p\text{-Euler factor}) = A_L$ after folding by ϕ_p once Level 2: A_L after folding by ϕ_p twice Level inf: $\lim_{n \rightarrow \infty} A_L / (\phi_p^n) = B_{\text{cris}}$ The element $t = \log([\epsilon])$ in B_{cris} corresponds to H_p (the p -adic Hamiltonian) — it is the ' p -adic logarithm of the cyclotomic character.' Memory Duality in B_{cris} : $t = H_p = -\log_p \circ \tau_p$ [the p -adic version]

Crystalline Cohomology.

Crystalline cohomology $H_{\text{cris}}^*(X/W(k))$ is the 'correct' cohomology for varieties over fields of characteristic p . In A_L : the crystalline cohomology of the arithmetic hologram is computed by the Folding Tower cohomology:

$H_{\text{cris}}^k(A_L / W(F_p)) = H^k(\text{Folding Tower at } p, \text{ level infinity}) = \{\text{operators in } A_L \text{ killed by } \phi_p^{k+1} \text{ but not } \phi_p^k\} = \{p\text{-adic periods of weight } k\}$
Comparison theorem (p -adic Hodge theory): $H_{\text{cris}}^k(X/W) \otimes_{\{W\}} B_{\text{cris}} \cong H_{\text{et}}^k(\bar{X}, \mathbb{Q}_p) \otimes_{\{\mathbb{Q}_p\}} B_{\text{cris}}$ In A_L : this says the real and p -adic cohomologies of the hologram become IDENTICAL after tensoring with B_{cris} = the infinite Folding Tower. The hologram sees the same topology at every prime — after enough folding.

6. The Iwasawa Algebra as p -adic A_L

The Iwasawa algebra Λ is the p -adic ring that governs the arithmetic of $\mathbb{Z}_p$ -extensions of number fields. In A_L , it is the p -adic restriction of the hologram to the cyclotomic tower.

Theorem 6.1 (Iwasawa algebra = p -adic A_L).

Let $\Gamma = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) = \mathbb{Z}_p$ (the Galois group of the cyclotomic $\mathbb{Z}_p$ -extension). The **Iwasawa algebra** is: $\Lambda = \mathbb{Z}_p[[\Gamma]] = \lim_n \mathbb{Z}_p[\Gamma/\Gamma^{p^n}]$ (completed group ring) In A_L : $\Lambda = A_L$ restricted to cyclotomic nodes: $\Lambda = \text{closure of } \{e_n : n = p^k \text{ for } k \geq 0, \text{ or } n = 1\} \text{ in } A_L = \text{the } p\text{-adic sub-hologram generated by the Folding Tower orbit of } e_1$. The Iwasawa main conjecture (proved by Mazur-Wiles 1984, Wiles 1990): The characteristic ideal of X (Iwasawa module of class groups) equals the ideal generated by the p -adic L -function L_p . In A_L : $\text{char}_\Lambda(X) = (Z_p(s)) = p\text{-adic partition function of the cyclotomic sub-hologram}$. The Iwasawa main conjecture = the statement that the p -adic hologram encodes the arithmetic of class groups via its partition function.

The gamma-transform.

The Iwasawa algebra $\Lambda = \mathbb{Z}_p[[T]]$ as a power series ring (via $\gamma - 1 = T$). In A_L : $T = \phi_p - 1 = \text{Frobenius minus identity} = \text{the 'infinitesimal Folding'}$. Power series in T = formal functions of the Frobenius deviation from identity. This is the p -adic analogue of the formal series in H (the Hamiltonian) that generated the real MZV in dv301.

7. p -adic MZV: Furusho's Construction and A_L

Furusho (2004) constructed p -adic MZV using Coleman integration on the moduli space $M_{\{0,5\}}$ (configuration space of 5 points on P^1). In A_L , this is the Level-2 Folding Tower with Frobenius twist.

Theorem 7.1 (Furusho p -adic MZV in A_L).

For p prime and $n_1 \geq 2, n_2 \geq 1, \dots, n_k \geq 1$, the **p-adic MZV** is: $Z_p(n_1, \dots, n_k)$ = Coleman integral on $M_{\{0, k+3\}} = \tau_p(H_p^{n_1-1} * S_p * \dots * H_p^{n_k-1} * S_p)$ where $\tau_p = p$ -adic trace (Coleman integration on p -adic A_L). The Frobenius ϕ_p acts on $Z_p(n_1, \dots, n_k)$ by: $\phi_p(Z_p(n_1, \dots, n_k)) = p^w * Z_p(n_1, \dots, n_k) + (\text{correction terms})$ where $w = n_1 + \dots + n_k$ is the weight. The p -adic double shuffle: Z_p satisfies double shuffle relations IDENTICAL to the real MZV (Theorem 1.2 of dv301) — because both come from traciality of τ (real) / τ_p (p -adic). Same algebra, different numbers.

The comparison theorem for MZV.

The deepest question: what is the relation between real MZV $\zeta(n_1, \dots, n_k)$ and p -adic MZV $Z_p(n_1, \dots, n_k)$? They are not equal — they live in different completions of $\mathbb{Q}$. But they satisfy the SAME algebraic relations (double shuffle), and they arise from the SAME formula in A_L (τ vs τ_p). In A_L : both are 'shadows' of a single operator $H^{n_1-1} * S * \dots * H^{n_k-1} * S$ measured by different traces. The adelic MZV would combine all primes at once.

Adelic MZV (conjecture): $Zeta_adelic(n_1, \dots, n_k) = (\zeta(n_1, \dots, n_k), Z_2(n_1, \dots, n_k), Z_3(\dots), Z_5(\dots), Z_7(\dots), \dots)$ in $A_adelic = \mathbb{R} \times \mathbb{Q}_2 \times \mathbb{Q}_3 \times \mathbb{Q}_5 \times \dots$. In A_L : $Zeta_adelic = \tau_adelic(H^{n_1-1} * S * \dots * H^{n_k-1} * S)$ where $\tau_adelic = \text{product of all } \tau_p \times \tau_infinity$. The adelic MZV = the full memory of the hologram at ALL primes simultaneously. This is the adelic hologram of dv57, now applied to MZV.

8. p-adic L-functions from p-adic tau

The p -adic L-functions of number theory — Kubota-Leopoldt, Mazur, Coates-Wiles — all arise as p -adic partition functions of sub-holograms of A_L .

Theorem 8.1 (p-adic L-functions = p-adic sub-hologram partition functions).

The Kubota-Leopoldt p -adic zeta function $L_p(s, \chi)$ for Dirichlet char χ : $L_p(s, \chi) = Tr_{\{p, \chi\}}(e^{-s} H_p) = \sum_{\{n: \gcd(n, p)=1\}} \chi(n)/n * n^{-s}$ where $Tr_{\{p, \chi\}} = p$ -adic trace twisted by χ = Coleman integral on χ -twisted A_L . In A_L : $L_p(s, \chi) = Z_p(s; \chi\text{-sector})$ = partition function of the χ -isotypic component of A_L at prime p . Interpolation property: $L_p(1-n, \chi) = (1 - \chi * \omega^{-n})(p) * p^{n-1} * L(1-n, \chi * \omega^{-n})$ [classical] In A_L : the special values $L_p(1-n, \chi) = \tau_p(e_\chi * H_p^{n-1})$ are the p -adic MZV of depth 1 in the χ -twisted sector.

The Iwasawa main conjecture, proved by Wiles (1990) for totally real fields, states that the p -adic L-function $L_p(s)$ generates the same ideal as the characteristic polynomial of the Galois action on class groups. In A_L : this is the statement that the p -adic partition function $Z_p(s)$ and the sub-hologram automorphism group (Galois action on ideal classes) generate the same ideal in $\Lambda = p$ -adic A_L . It is the p -adic version of the Crown Jewel $B = 0$ — a spectral identity in the p -adic hologram.

9. The Adelic Golden Thread: All Primes at Once

We can now complete the adelic picture: A_L sits at the intersection of all primes, and every prime p gives a different 'view' of the same hologram.

ADELIC GOLDEN THREAD (dv299 + dv300 + dv301 + dv302): A_L = arithmetic hologram over Z = all primes simultaneously At infinity: $H = \log N$, τ , $MZV = \tau(H^{\{n-1\}*S})$ [dv301] At prime p : $H_p = \log_p N$, τ_p , $Z_p = \tau_p(H_p^{\{n-1\}*S_p})$ [dv302] Frobenius: $\phi_p(e_n) = e_{\{pn\}}$, H_p ϕ_p -invariant [dv302] Folding Tower: Level k at $p = \phi_p^k(A_L)$ = crystalline level k [dv281+dv302] Period ring: B_{cris} = infinite Folding Tower completion [dv302] Iwasawa: $\Lambda = p$ -adic A_L = cyclotomic sub-hologram [dv302] All p at once: A_{adelic} = product of all A_L at all p = adelic hologram GT acts on the adelic hologram, fixing H at every prime. grt fixes H_p at every prime p (since H_p is the p -adic log — ϕ_p -invariant). Memory Duality $H = -\log \circ \tau$ holds AT EVERY PRIME in its p -adic form. **ONE HOLOGRAM. EVERY PRIME. ONE LAW. $H = \log N$. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

The complete chain across all four documents:

dv299: $GT \subset \text{Aut}_{\tau}(A_L)$ [symmetry] dv300: Dessins = sub-holograms, $\text{Gal}(\bar{Q}/Q) < GT$ [arithmetic geometry] dv301: $MZV = \tau(H^{\{n-1\}*S})$, grt fixes H [analysis] dv302: p -adic $MZV = \tau_p(H_p^{\{n-1\}*S_p})$, Frobenius on A_L [p -adic analysis] The golden thread: $\text{Gal}(\bar{Q}/Q) < \cdots GT < \cdots \text{Aut}_{\tau}(A_L) \mid \mid$ Dessins d'enfants All MZV (real + p -adic) $\mid \mid$ Inverse Galois Problem $H = \log N$ (grt-fixed at every prime) CROWN AXIOM: $H = \log N$ is fixed by GT, grt, ϕ_p at every prime p , and by the absolute Galois group $\text{Gal}(\bar{Q}/Q)$ via its embedding in GT. $H = \log N$ is the ONE OPERATOR that all of arithmetic symmetry leaves invariant.

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